

Neural Decoding

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- Encoding and Decoding
- Discrimination
- Population Decoding

Review for the basic

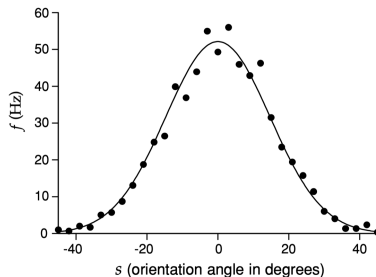
Average **firing rate** is

$$\langle r \rangle = \frac{\langle n \rangle}{T} = \frac{1}{T} \int_0^T d\tau \langle \rho(\tau) \rangle = \frac{1}{T} \int_0^T dt r(t),$$

where $\rho(t)$ is a response function. Consider it as a function of s ,

$$\langle r \rangle = f(s),$$

which is called the neural response **tuning curve**.



Gallery of notation

Throughout this presentation...

- s : stimulus,
- $\mathbf{r} = (r_1, \dots, r_N)$: a list of spike-count firing rates for N neurons,
- $P[s]$: the probability of stimulus s being presented,
- $P[\mathbf{r}]$: the probability of response $\mathbf{r}$ being recorded,
- $P[\mathbf{r}, s]$: the probability of stimulus s being presented and response $\mathbf{r}$ being recorded,
- $P[\mathbf{r}|s]$: the conditional probability of evoking response $\mathbf{r}$, given that stimulus s was presented,
- $P[s|\mathbf{r}]$: the conditional probability that stimulus s was presented, given that response $\mathbf{r}$ was recorded.

Basic Probability

From

$$P[r, s] = P[r|s]P[s] = P[s|r]P[r],$$

we derive the **Bayes theorem**,

$$P[s|r] = \frac{P[r|s]P[s]}{P[r]}, \quad \text{when } P[r] \neq 0.$$



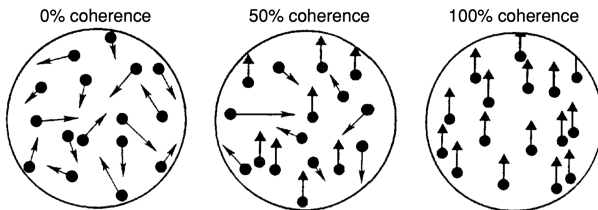
Figure: Thomas Bayes

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Experiment

Task : Report the **direction of motion** in the random dot images.



Up($\uparrow$) : plus direction(+), Down($\downarrow$) : minus direction(-).

Experiment Result

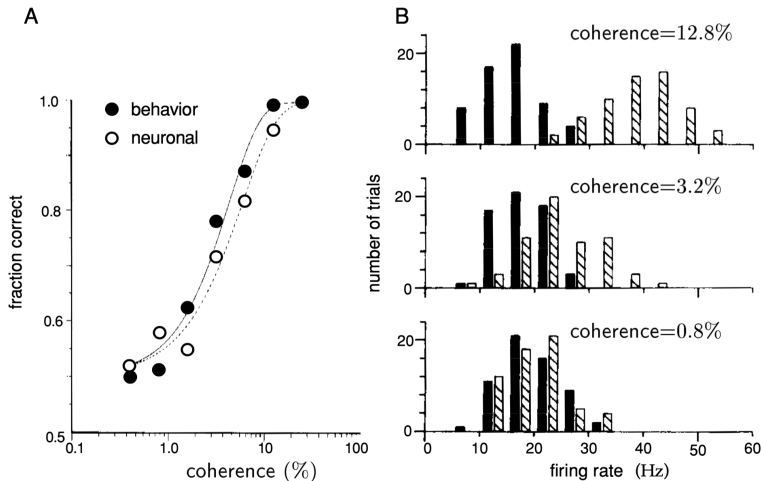


Figure: Behavioral and electrophysiological results

Discrimination

Approximate the two distributions as

Gaussian $N(\langle r \rangle_+, \sigma_r^2)$ and $N(\langle r \rangle_-, \sigma_r^2)$.

We use a measure of the [separation](#)

$$d' = \frac{\langle r \rangle_+ - \langle r \rangle_-}{\sigma_r} \quad : \quad \text{discriminability.}$$

Let's consider a [threshold](#) number z such that

If $r \geq z$, report "plus",
if $r < z$, report "minus".

Discrimination

Define

$\alpha(z) := P[r \geq z | -]$: false alarm of the test,

$\beta(z) := P[r \geq z | +]$: hit rate of the test.

stimulus	probability	
	correct	incorrect
+	β	$1 - \beta$
-	$1 - \alpha$	α

Want to find z s.t. $\alpha(z) \approx 0$ and $\beta(z) \approx 1$.

ROC Curves

ROC curve = **R**eceiver **O**perating **C**haracteristic curve

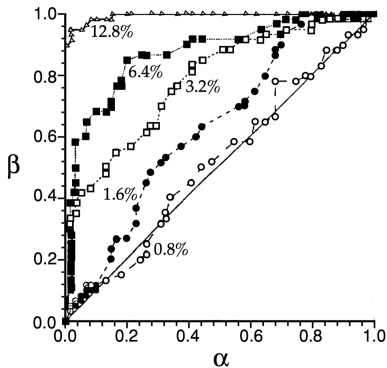


Figure: Behavioral and electrophysiological results

ROC Curves

Consider

$$P[\text{correct}] = \int_0^\infty dz p[z|-] P[r \geq z|+] = \int_0^\infty dz p[z|-] \beta(z).$$

By using

$$\alpha(z) = \int_z^\infty dr p[r|-], \quad \text{hence,} \quad \frac{d\alpha}{dz} = -p[z|-],$$

we have

$$P[\text{correct}] = \int_0^1 d\alpha \beta = (\text{area under the ROC curve}).$$

Likelihood Ratio Test

By Neyman-Pearson lemma, it is optimal to use

$$l(r) = \frac{p[r|+]}{p[r|-]}.$$

Consider a loss parameter

L_+ : answering “minus” when the correct is “plus”,

L_- : answering “plus” when the correct is “minus”.

Using Bayes theorem,

$$P[+|r] = \frac{p[r|+]P[+]}{p[r]} \quad \text{and} \quad P[-|r] = \frac{p[r|-]P[-]}{p[r]}.$$

Likelihood Ratio Test

- 1 Relation between the **likelihood ratio** and the **ROC curve**.

By definition of $\beta(z)$,

$$\beta(z) = \int_z^\infty dr p[r|+] \quad \text{so} \quad \frac{d\beta}{dz} = -p[z|+],$$

which derives

$$\frac{d\beta}{d\alpha} = \frac{d\beta}{dz} \frac{dz}{d\alpha} = \frac{p[z|+]}{p[z|-]} = l(z).$$

Likelihood Ratio Test

- ② Comparing the likelihood ratio to a threshold values is an optimal decoding procedure for discrimination.

Set

$$\text{Loss}_+ = L_+ P[-|r] \quad \text{and} \quad \text{Loss}_- = L_- P[+|r].$$

Strategy Answer “plus” if $\text{Loss}_+ \leq \text{Loss}_-$ (and vice versa).

This gives

$$l(r) = \frac{p[r|+]}{p[r|-]} \geq \frac{L_+}{L_-} \frac{P[-]}{P[+]},$$

obtained from the Bayes theorem.

Likelihood Ratio Test

If we discriminate between $s + \Delta s$ and s ,

$$\frac{p[r|s + \Delta s]}{p[r|s]} \approx \frac{p[r|s] + \Delta s \partial p[r|s] / \partial s}{p[r|s]} = 1 + \Delta s \frac{\partial \ln p[r|s]}{\partial s}.$$

Hence, for small Δs , the likelihood ratio test is equivalent to compare

$$Z(r) = \frac{\partial \ln p[r|s]}{\partial s}$$

to a threshold $(z - 1) / \Delta s$. The function $Z(r)$ is called the **score**.

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Vector method

Topic : cercal system of the cricket.

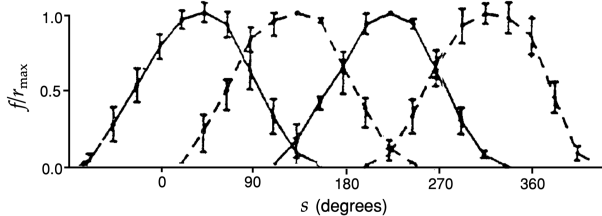


Figure: Tuning curves for the four interneurons of the cricket cercal system

Vector method

Tuning curve for interneuron a ($a = 1, 2, 3, 4$) is written as

$$\left(\frac{f(s)}{r_{max}} \right)_a = [\cos(s - s_a)]_+ = [\vec{v} \cdot \vec{c}_a]_+.$$

From this, we obtain the direction

$$\vec{v}_{pop} = \sum_{a=1}^4 \left(\frac{r}{r_{max}} \right)_a \vec{c}_a.$$

Bayesian inference

Idea : **Minimize** a loss function $L(s, s_{\text{bayes}})$, i.e.,

$$\min_{s_{\text{bayes}}} \int ds L(s, s_{\text{bayes}}) p[s|\mathbf{r}].$$

e.g. If $L(s, s_{\text{bayes}}) = (s - s_{\text{bayes}})^2$, then,

$$s_{\text{bayes}} = \int ds p[s|\mathbf{r}] s.$$

MAP inference

MAP inference = **M**aximum **A** **P**osteriori inference

Idea : Choose the stimulus s_{MAP} that maximize the conditional probability density of the stimulus, i.e.,

$$s_{MAP} = \arg \max_s p[s|r].$$

If $p[s]$ is independent of s , then, $p[s|r]$ and $p[r|s]$ have the same dependence on s . So, MAP is equivalent to maximizing the likelihood function, i.e.,

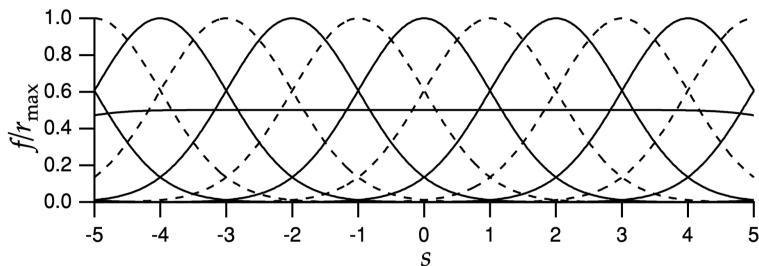
$$s_{ML} = \arg \max_s p[r|s].$$

We call it by maximum likelihood (ML) inference.

Example

Consider Gaussian tuning curves

$$f_a(s) = r_{\max} \exp \left(-\frac{1}{2} \left(\frac{s - s_a}{\sigma_a} \right)^2 \right), \quad a = 1, \dots, N.$$



Example

Assume (1) firing rate r_a is determined by n_a/T . (2) variability can be described by the **homogeneous Poisson model**. Then, we have (Chapter 1)

$$P[r_a|s] = \frac{(f_a(s)T)^{r_a T}}{(r_a T)!} \exp(-f_a(s)T).$$

Under assumption of **independence** of neuron fires,

$$P[r|s] = \prod_{a=1}^N \frac{(f_a(s)T)^{r_a T}}{(r_a T)!} \exp(-f_a(s)T).$$

Example (ML algorithm)

For ML algorithm, we only consider the term depending on s ,

$$\ln P[\mathbf{r}|s] = T \sum_{a=1}^N r_a \ln(f_a(s)) + \dots$$

To find s_{ML} which maximizes $\ln P[\mathbf{r}|s]$,

$$\sum_{a=1}^N r_a \frac{f'_a(s_{ML})}{f_a(s_{ML})} = \sum_{a=1}^N r_a \frac{s_a - s_{ML}}{\sigma_a^2} = 0,$$

hence,

$$s_{ML} = \frac{\sum r_a s_a / \sigma_a^2}{\sum r_a / \sigma_a^2}.$$

Example (MAP algorithm)

By Bayes theorem,

$$\ln p[s|\mathbf{r}] = \ln P[\mathbf{r}|s] + \ln p[s] - \ln P[\mathbf{r}].$$

Then, only considering the s -dependent terms,

$$\ln p[s|\mathbf{r}] = T \sum_{a=1}^N r_a \ln(f_a(s)) + \ln p[s] + \dots$$

To find s_{MAP} which maximizes $\ln p[s|\mathbf{r}]$,

$$T \sum_{a=1}^N \frac{r_a f'_a(s_{MAP})}{f_a(s_{MAP})} + \frac{p'[s_{MAP}]}{p[s_{MAP}]} = 0,$$

which deduces

$$s_{MAP} = \frac{T \sum r_a s_a / \sigma_a^2 + s_{prior} / \sigma_{prior}^2}{T \sum r_a / \sigma_a^2 + 1 / \sigma_{prior}^2}.$$

Accuracy

We evaluate the accuracy by using

$$b_{est}(s) = \langle s_{est} \rangle - s \quad \text{and} \quad \sigma_{est}^2(s) = \langle (s_{est} - \langle s_{est} \rangle)^2 \rangle.$$

An estimate is said to be **unbiased** if $b_{est}(s) = 0$ for all s .

Note that

$$\langle (s_{est} - s)^2 \rangle = \langle (s_{est} - \langle s_{est} \rangle + b_{est}(s))^2 \rangle = \sigma_{est}^2(s) + b_{est}^2(s).$$

Fisher Information

The **Cramer-Rao bound** limits the variance of any estimate s_{est} as

$$\sigma_{est}^2(s) \geq \frac{(1 + b'_{est}(s))^2}{I_F(s)},$$

where $I_F(s)$ is the **Fisher information** of the firing-rate distribution. In detail,

$$\begin{aligned} I_F(s) &= \left\langle -\frac{\partial^2 \ln p[\mathbf{r}|s]}{\partial s^2} \right\rangle \\ &= \left\langle \left(\frac{\partial \ln p[\mathbf{r}|s]}{\partial s} \right)^2 \right\rangle = \int d\mathbf{r} p[\mathbf{r}|s] \left(\frac{\partial \ln p[\mathbf{r}|s]}{\partial s} \right)^2. \end{aligned}$$

Fisher Information

If the estimate of the stimulus is **unbiased**, i.e., $b_{est}(s) = 0$, then,

$$\sigma_{est}^2 \geq \frac{1}{I_F(s)}.$$

In the limit of **large numbers** of encoding neurons, and for most firing-rate distributions, the ML estimate is unbiased and saturates the Cramer-Rao bound, i.e.,

$$\sigma_{ML}^2(s) = \frac{1}{I_F(s)}.$$

We call such an estimator to be **efficient**.

Thanks for listening.